Bandi Sharanmegha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.46
Harvest Public School	Class XII (CBSE)	2024	81.20%
Resonance Whiz School	Class X (SSC)	2022	9.80%

PROJECTS

Apex-Gateway | Distributed API Gateway & Rate Limiter

- Engineered a reverse-proxy API gateway in Go supporting dynamic route mapping, JWT/IP-based authentication, backend health checks, and hot-config reloading without process restarts.

- Developed thread-safe Token Bucket and Sliding Window rate limiters; verified zero over-admission under high mutex-contention testing while achieving sub-microsecond core evaluation latency.

- Designed a two-tier (L1 Local Memory / L2 Redis) caching architecture using atomic Redis Lua scripts to handle node synchronization, backed by a circuit breaker for fault-tolerant degradation.

- Containerized a 3-node gateway cluster behind NGINX with Docker Compose and authored k6/Vegeta load-testing suites to benchmark end-to-end cluster throughput and network latency.

xv6-RISCV Extensions | Kernel Systems & Memory Management

- Extended the MIT xv6-RISCV kernel (~870 LOC) with custom trap handlers, store-page-fault interception, and kernel-scheduled thread primitives sharing a single address space (NPROC=64 limit).

- Implemented Copy-on-Write (COW) fork mechanics via physical page reference counting and custom PTE software bits, deferring page allocation to first-write to minimize process initialization overhead.

- Built a file-backed mmap() subsystem featuring lazy demand-paging on page faults, dirty-page write-back, and a Clock (second-chance) page eviction algorithm utilizing hardware PTE_A bits.

- Designed a user-space threading library with clone/join syscalls and atomic (amoswap) spinlocks, enabling lightweight, kernel-scheduled concurrency within shared virtual memory.

CUDA 2D Numerical Solver | High-Performance Computing & Modern C++

- Engineered a parallel 2D Finite Difference advection-diffusion PDE solver in C++ and CUDA, processing millions of grid nodes with continuous 1D spatial memory layouts.

- Accelerated core stencil updates via CUDA shared-memory kernels, achieving a 28.7x speedup (96.5% compute latency reduction) over optimized C++ on an NVIDIA T4 GPU.

- Reached peak GPU throughput of 4.27 Billion cell-updates/sec for 500-timestep simulations, validating numerical convergence against analytical decay solutions.

- Integrated OpenMP multi-threading and exported VTK spatial datasets for real-time fluid dynamic visualizations in ParaView.

SKILLS

Programming Languages: C++, C, Go, Python, RISC-V Assembly, CUDA C++

Developer Tools & Infrastructure: Git, Docker, Docker Compose, NGINX, CMake, GDB, QEMU, Linux/POSIX

Frameworks & Tech Stack: Redis, OpenMP, gRPC/REST APIs, k6, Vegeta

Libraries & Visualization: NumPy, Pandas, Matplotlib, ParaView (VTK)

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Operating Systems

RELEVANT COURSES

Computer Science Courses - Introduction to Programming | Data Structures & Applications | Operating Systems

Computational Courses - Computer Aided Numerical Methods | Fundamentals of Scientific Computing | Computational Methods in Material Science | Electronic Structure and Atomistic Modeling of Materials

Math Courses - Calculus | Linear Algebra | Differential Equations | Intro to Probability | Transform Techniques | Introduction to Statistics | Complex Variables

Entrepreneurship Courses - Intro to Entrepreneurship | Fundamentals of Organizational Structure

POSITIONS OF RESPONSIBILITY

Overall Coordinator | Abhyanth, IIT Hyderabad Sports Fest (2026-27) -

Leading the planning and execution of the institute's annual sports fest, overseeing multiple domains, coordinators and volunteers to ensure seamless event operations.

Hospitality Coordinator | Elan&nVision (2025-56) -

Managed hospitality and logistics for guests, participants, and dignitaries during the institute's annual techno-cultural fest.

Operations Coordinator | International Relations (IR) Cell (2025-26) -

Assisted international exchange students and coordinated operational activities for international programs and events.

Editorial Coordinator | Media Council (2025-26) -

Led editorial activities by creating, reviewing, and publishing institute magazines and written content.

Outreach Coordinator | Alumni Cell (2025-26) -

Facilitated alumni engagement through outreach initiatives and coordination for institute events.

EXTRACURRICULAR

- Member, **National Sports Organization (NSO) - Athletics**, IIT Hyderabad
- Badminton, Table Tennis, Running